ABL PYTHON PROGRAMMING LABORATORY MANUAL

ACTIVITY-BASED LEARNING COURSE CODE & EXECUTION WORKBOOK

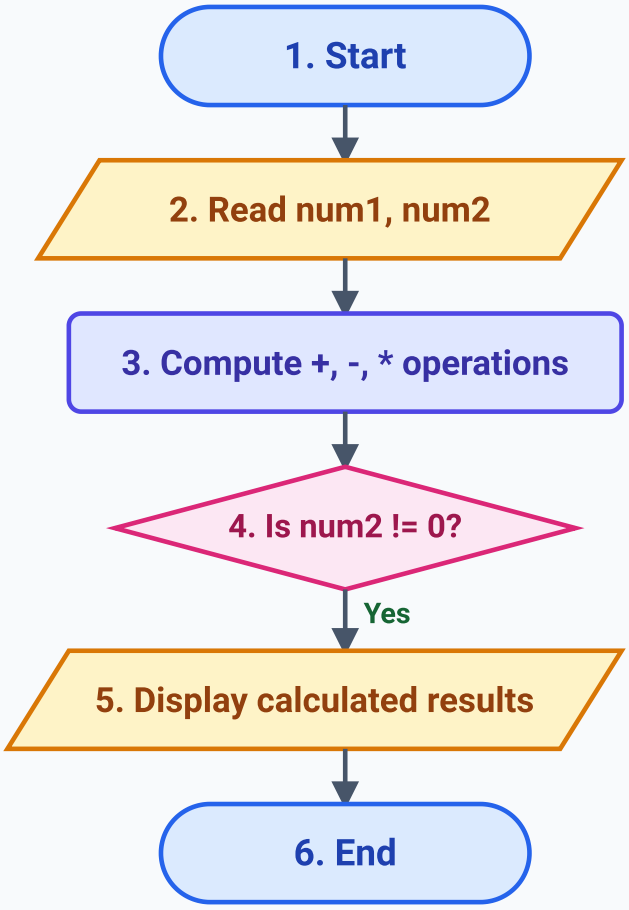
Course Title	Activity-Based Learning: Python Programming	**Document Code**	ABL-PY-2026-MANUAL
Target System	Python 3.x Interpreter / IDLE / Jupyter	**Standard Format**	Aim | Algorithm | Flowchart | Pseudo Code | Code | Output | Conclusion

EXPERIMENT 01: Arithmetic Operations & Data Input — LAB SEC: ABL-01

1. AIM

To write and execute a Python program that accepts two numeric inputs from the user and performs basic arithmetic operations (Addition, Subtraction, Multiplication, Division, and Modulus).

2. FLOWCHART

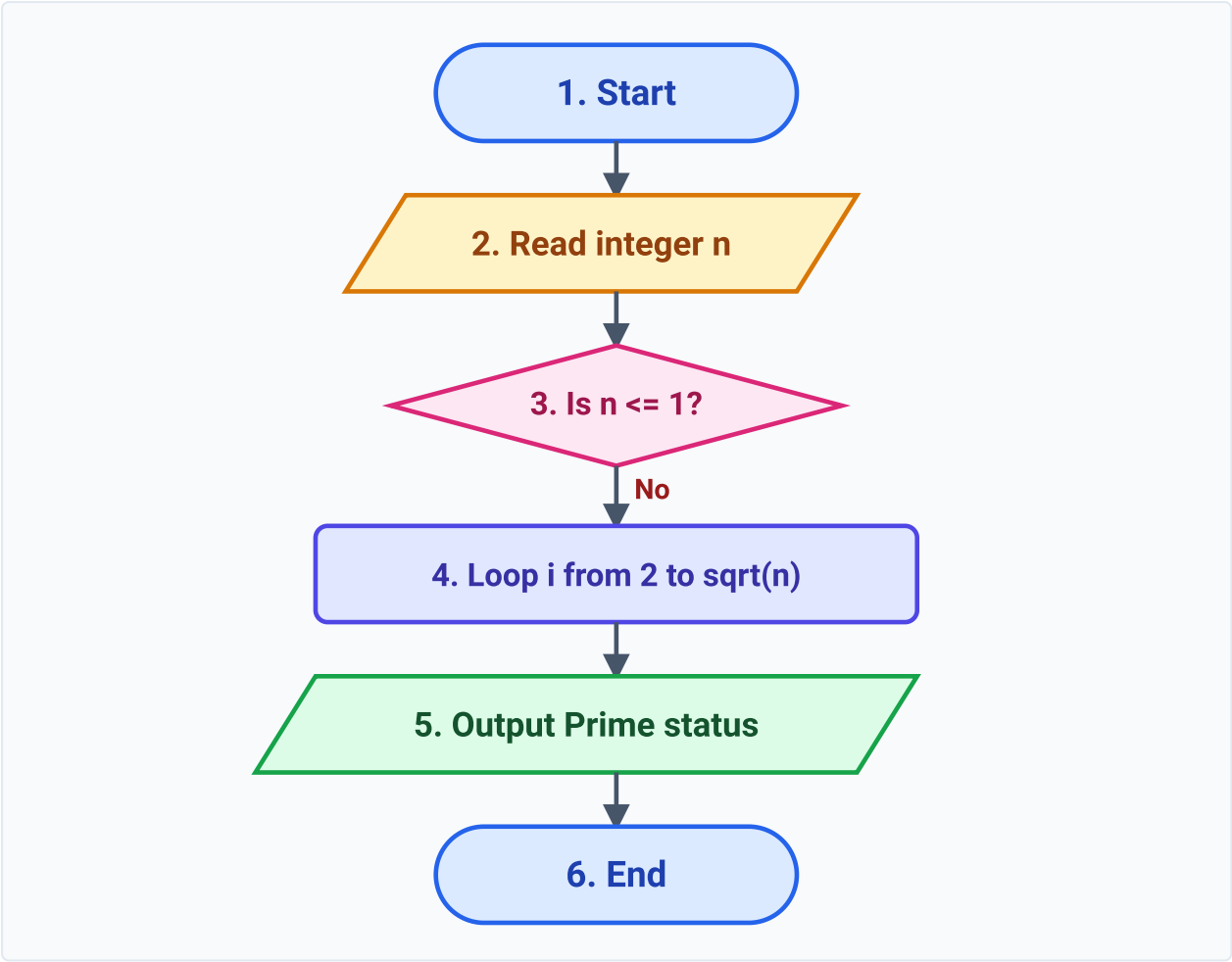


3. ALGORITHM

1. AIM

To design a Python program to check whether a given integer is a prime number using conditional statements and loops.

2. FLOWCHART

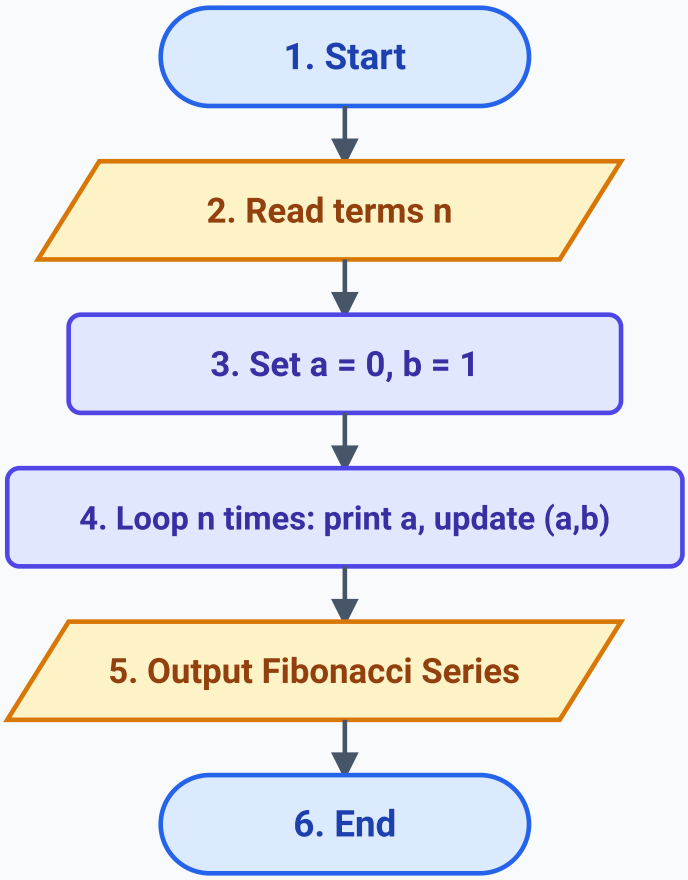


3. ALGORITHM

1. AIM

To develop a Python program that generates and prints the Fibonacci series up to n terms using loop structures.

2. FLOWCHART

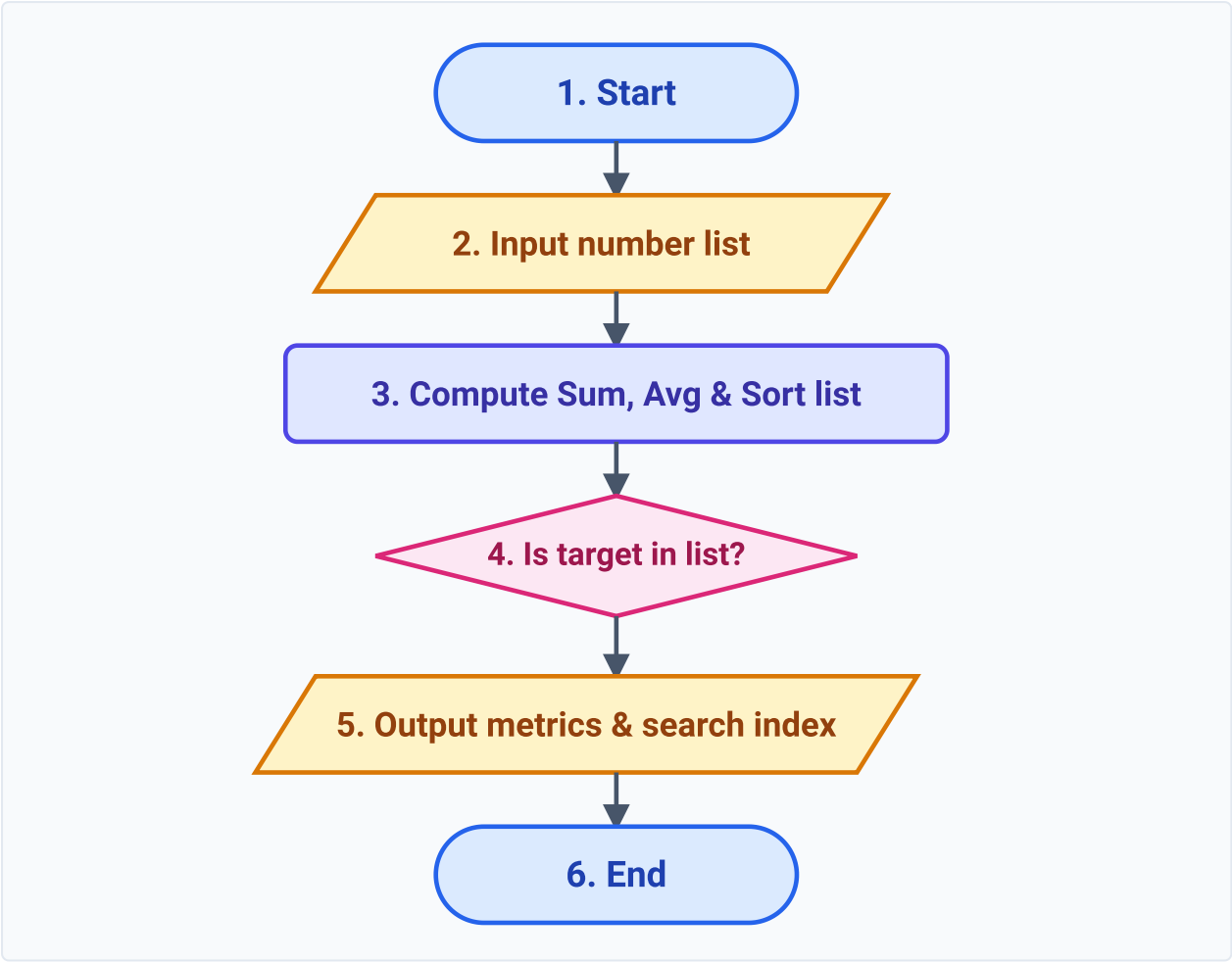


3. ALGORITHM

1. AIM

To implement list manipulations in Python including insertion, element search, calculation of sum/average, and sorting.

2. FLOWCHART

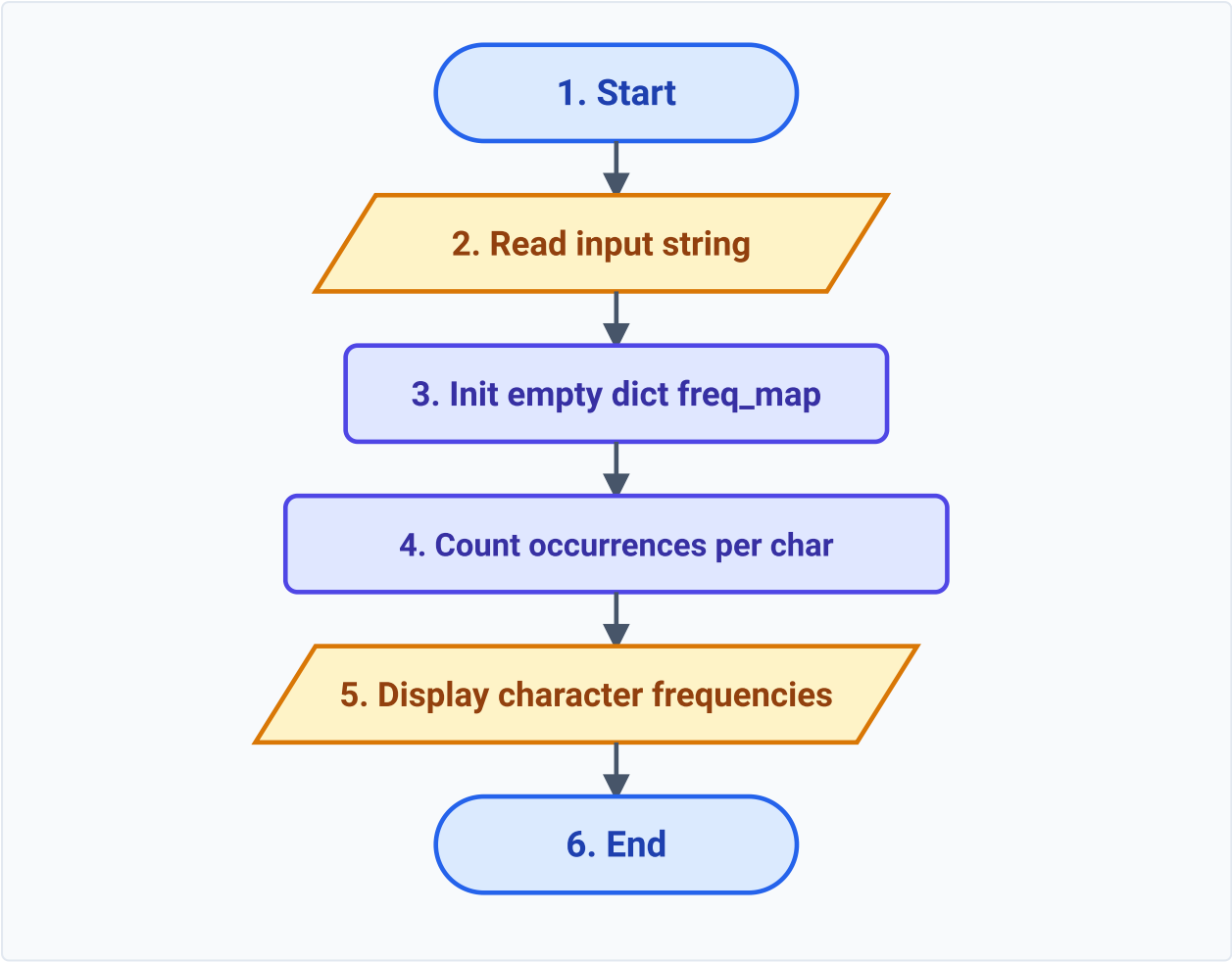


3. ALGORITHM

1. AIM

To construct a Python program that accepts a string input and computes the occurrence frequency of each unique character using dictionary data structures.

2. FLOWCHART

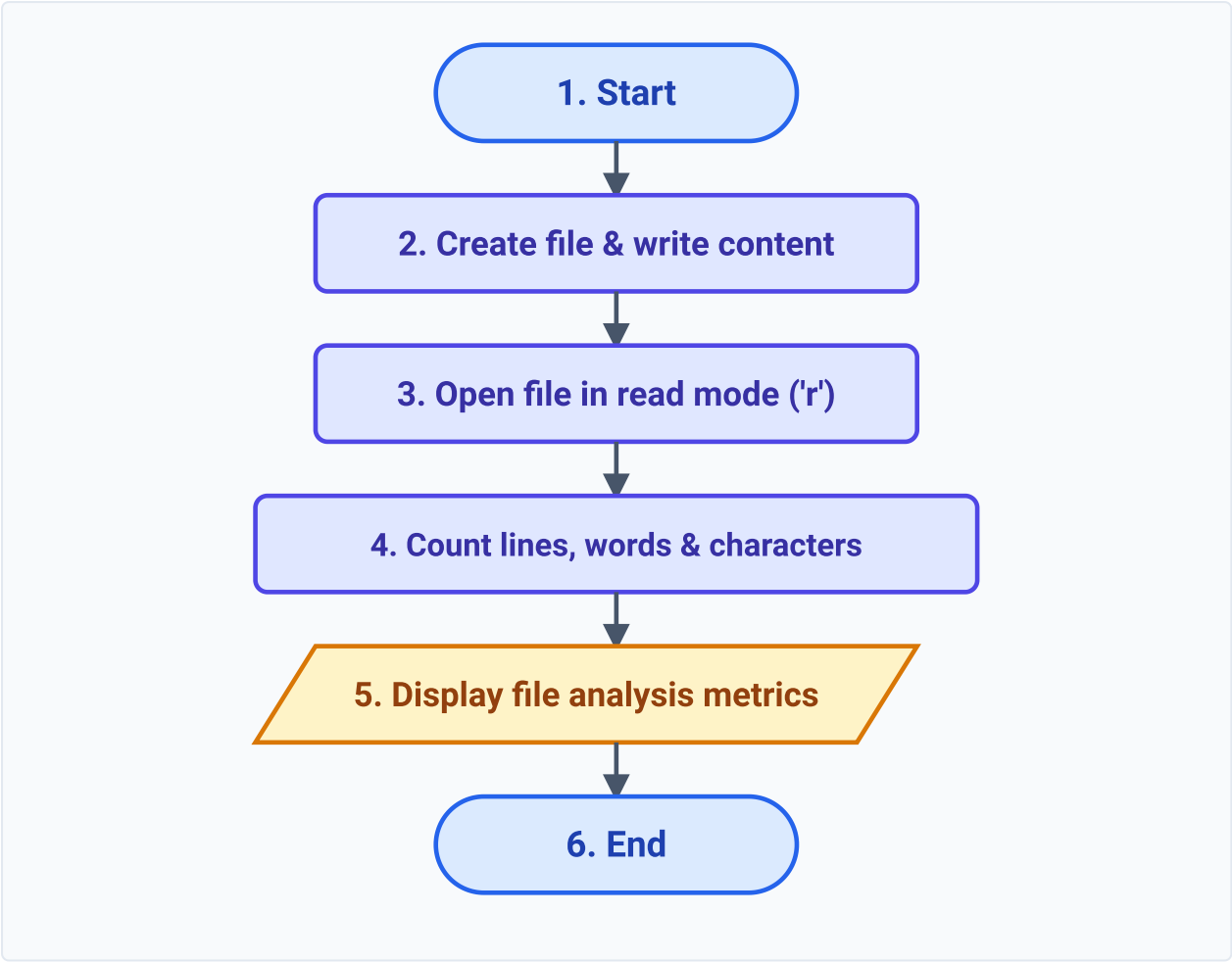


3. ALGORITHM

1. AIM

To write a Python program that creates a text file, writes content into it, reads the file back, and counts the total number of lines, words, and characters.

2. FLOWCHART


1. Start
2. Create file & write content
3. Open file in read mode ('r')
4. Count lines, words & characters
5. Display file analysis metrics
6. End

3. ALGORITHM